

MODELING PIPELINE

1. Choose a model

How should we represent the world?

2. Choose a loss function

How do we quantify prediction error?

3. Fit the model

How do we choose the best parameters of our model given our data?

4. Evaluate model performance

How do we evaluate whether this process gave rise to a good model?

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Let's start from the end and first focus here.

EVALUATING MODEL PERFORMANCE

What to compare against?

Baseline Models: Simplest reasonable models to compare against

EVALUATING MODEL PERFORMANCE

Possible Baselines for Classification:

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Features				Output	
x_{11}	x_{12}	$\dots$	x_{1p}	Yes	Training data
x_{21}	x_{22}	$\dots$	x_{2p}	No	
$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$	
$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$	Test data
x_{n1}	x_{n2}	$\dots$	x_{np}	Yes	

Don't care about the training data.

Assign 'Yes' to each row's label.

EVALUATING MODEL PERFORMANCE

Possible Baselines for Classification:

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e.g. assign “yes” or “no” randomly for each row.

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Find most common label. Say it is 'No'

Assign 'No' to each row's label

EVALUATING MODEL PERFORMANCE

Possible Baselines for Classification:

- Other baselines:
 - The best performing single-variable model.
 - Same label as the previous point in time:
Time series forecasting. e.g. did it rain yesterday?
 - A model built by someone else

EVALUATING MODEL PERFORMANCE

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Features

Output

x_{11}	x_{12}	$\dots$	x_{1p}	3.2
x_{21}	x_{22}	$\dots$	x_{2p}	4.0
$\vdots$	$\vdots$	$\cdot$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\cdot$	$\vdots$	$\vdots$
x_{n1}	x_{n2}	$\dots$	x_{np}	3.6

Training data

Test data

Find mean or median

Ignore features, assign mean or median to each row

EVALUATING MODEL PERFORMANCE

Possible Baselines for Prediction (Regression):

- Other baselines:
 - Linear Regression: Details next week.
 - Same value as the previous point in time:
Time series data: e.g. What was temperature yesterday?

EVALUATING CLASSIFIERS

Confusion matrix (contingency table) in a binary classifier:

		Predicted Class	
		Yes	No
Actual Class	Yes	TP	FN
	No	FP	TN

True positives (TP): + is labeled +

True negative (TN): - is labeled -

False positives (FP): - is labeled +

False negatives (FN): + is labeled -

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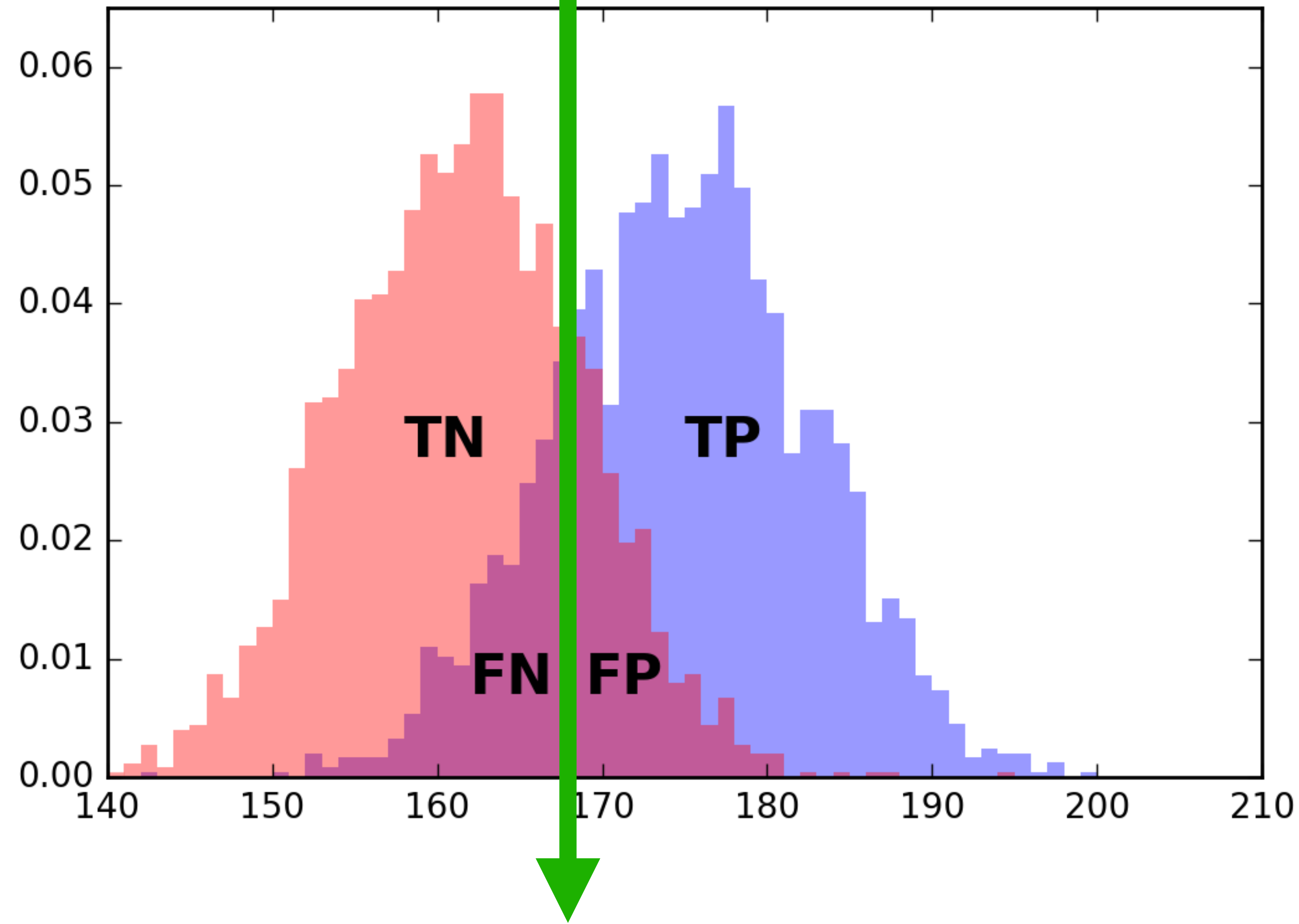
False negatives (FN): + is labeled -

Choose more interesting class as **positive**.

Example: Spam classification $\Rightarrow$ Spam positive, non-spam negative.

EVALUATING CLASSIFIERS

Example: Classify everyone of height ≥ 168 centimeters as male



MEASURES FOR EVALUATING CLASSIFIERS

Accuracy: Ratio of correct predictions over all predictions.

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What happens when the classes are unbalanced?

Example: Cancer diagnosis. Positive class (cancer) 5 % .

Blind classifier accuracy: 50 %

Most common label classifier accuracy: 95 %

MEASURES FOR EVALUATING CLASSIFIERS

Accuracy: Ratio of correct predictions over all predictions.

$$accuracy = \frac{TP + TN}{TP + FP + TN + FN}$$

Accuracy can be misleading if class imbalance.

We need measures sensitive to getting more TP.

MEASURES FOR EVALUATING CLASSIFIERS

Precision: Ratio of correct positive predictions over positive predictions.

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The case with unbalanced classes (problematic for accuracy)

Example: Cancer diagnosis. Positive class (cancer) 5 % .

Blind classifier precision: 5 %

Most common label classifier precision: undefined

Least common label classifier precision: 5 %

MEASURES FOR EVALUATING CLASSIFIERS

Recall: Ratio of correct positive predictions over all positive instances.

$$recall = \frac{TP}{P} = \frac{TP}{TP + FN}$$

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Example: Cancer diagnosis. Positive class (cancer) 5 % .

Blind classifier recall: 50 %

Most common label classifier recall: 0 %

Least common label classifier recall: 100 %

MEASURES FOR EVALUATING CLASSIFIERS

Precision vs Recall:

$$precision = \frac{TP}{TP + FP}$$

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Precision vs Recall:

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Are False Positives more important or False Negatives?

Preliminary diagnoses tests for cancer:

Might be ok to have FPs but not ok to have FNs.

Spam detection:

Might be ok to have FNs but not ok to have FPs.

MEASURES FOR EVALUATING CLASSIFIERS

F-score: Harmonic mean of precision and recall.

$$F = 2 \times \frac{\textit{precision} \times \textit{recall}}{\textit{precision} + \textit{recall}}$$

For a good F-score both precision and recall must be good.

MEASURES FOR EVALUATING CLASSIFIERS

Summary:

- Class sizes substantially different $\Rightarrow$ accuracy is misleading.
- High precision is hard to achieve in unbalanced class sizes:
If classifier makes many positive predictions, most will miss the mark
If classifier makes few positive predictions, few will catch rare positives
- F-score does the best job of any single statistic, but all four work together to describe the performance of a classifier.

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ROC (Receiver Operating Characteristic) Curve:

Class prediction usually via some 'score' reflecting **in classness**.

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Increasing height

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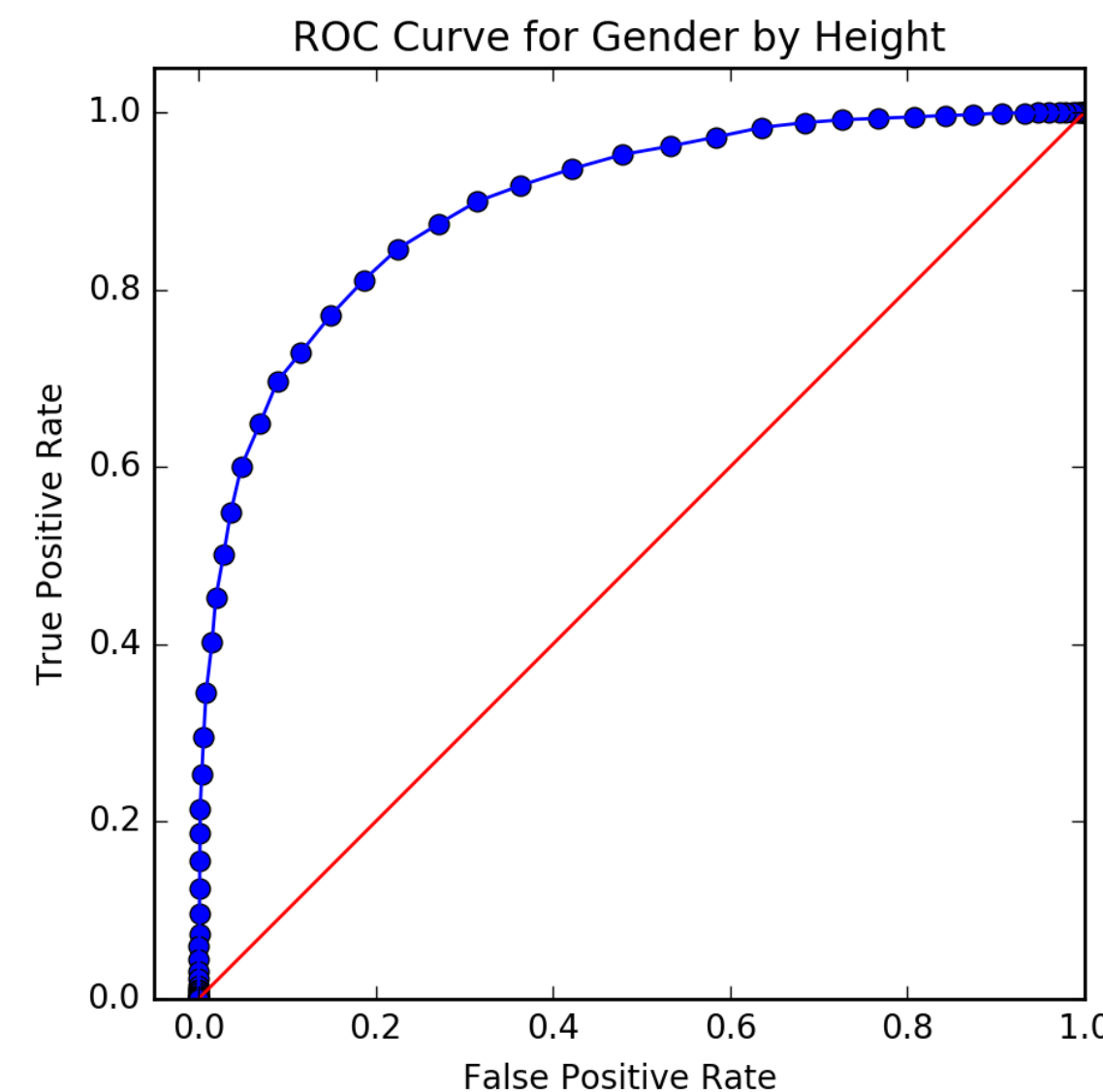
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For increasing values of height threshold:

$$TPR = Recall = \frac{TP}{P}$$

$$FPR = \frac{FP}{N}$$

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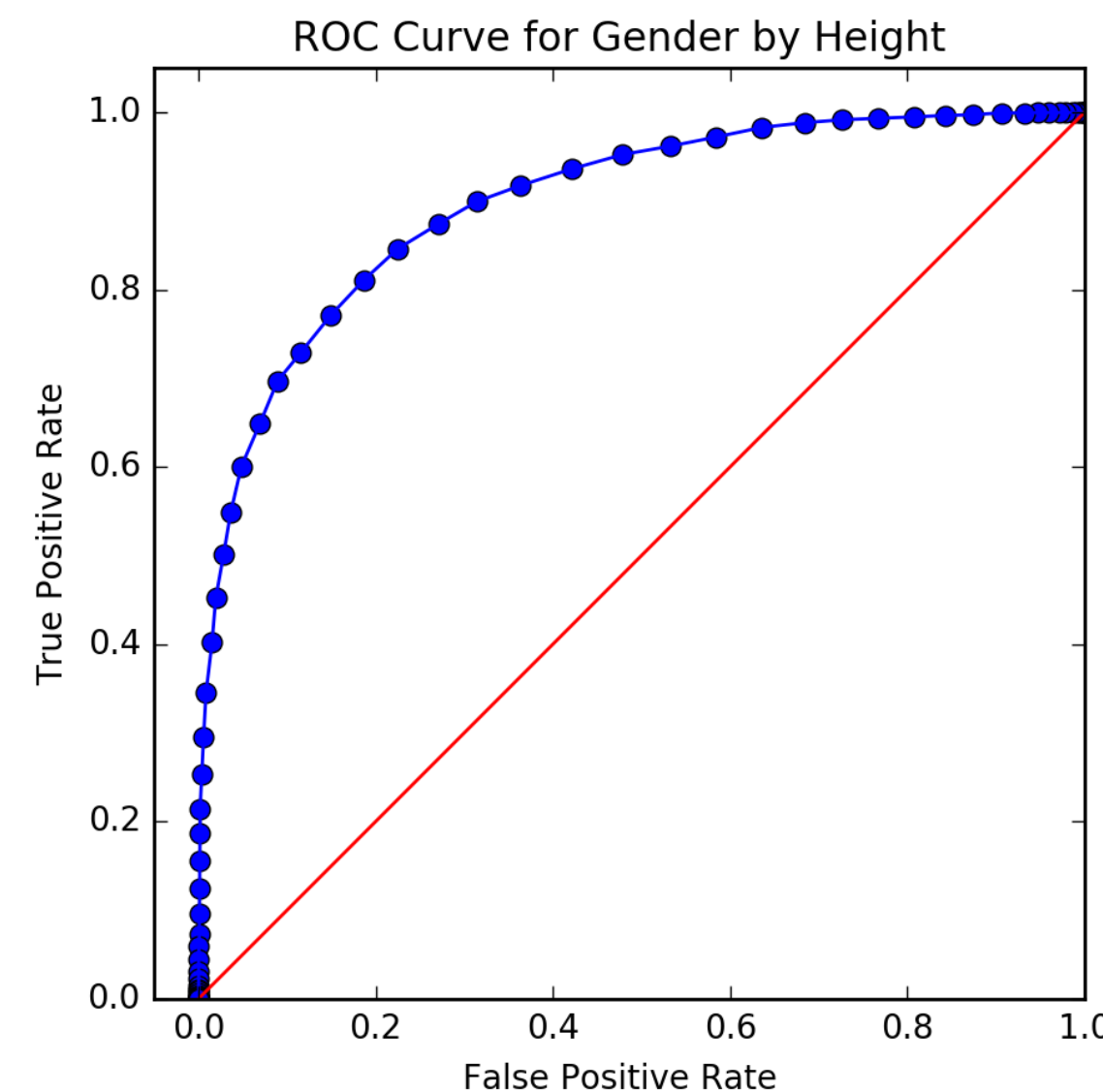
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How good is the score defining the classification.

At most 1.

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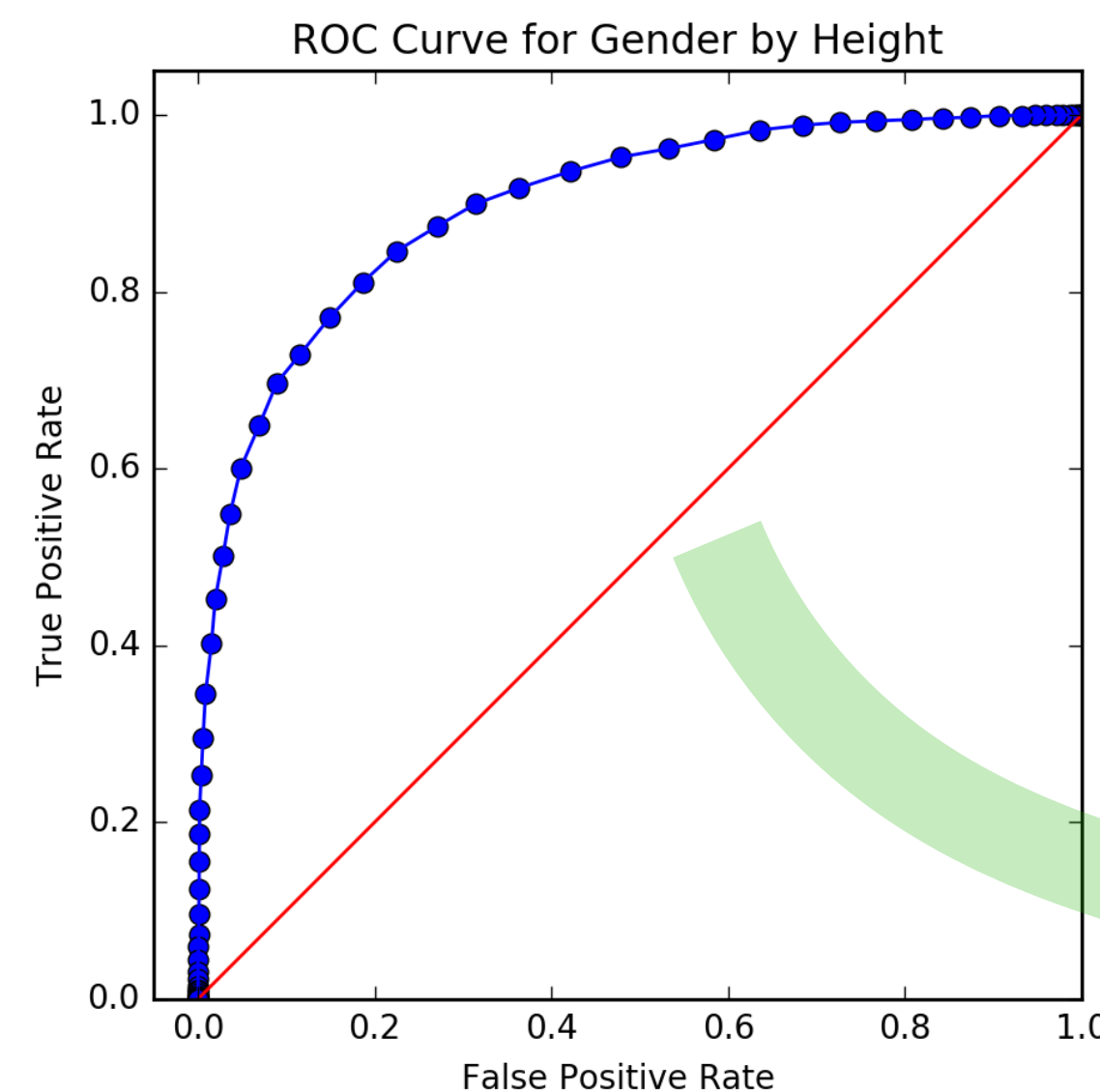
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Diagonal: Blind classifier.

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